Timothy Jin

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EDUCATION

The University of British Columbia | *B.Sc. Computer Science — 3.9 / 4.0 GPA* Sep 2021 – May 2026

Distributed Systems · Data Structures & Algorithms · Machine Learning · Networking & Security · Databases

EXPERIENCE

Amazon | *Software Development Engineer Intern* May 2025 – Aug 2025

- Eliminated engineer intervention for 80% of tickets submitted by internal teams requesting help with the Prime Subscriptions API by building an **agentic LLM** application powered by an **AWS** serverless backend in **Java, Python, and TypeScript** to automatically analyze user issues, surface insights, and run API tests.
- Reduced debugging time by 80% by migrating the Prime Subscriptions API from compressed file-based logs (requiring manual grepping) to centralized, queryable CloudWatch logging across 9 global deployments.
- Reduced manual scaling work by 95% for high-traffic shopping events (e.g. Prime Day) by integrating the Prime API into an automated capacity planning interface, replacing a process that previously required weeks of manual effort.

Stanford University - Dept. of Emergency Medicine | *Machine Learning Researcher* May 2024 – Oct 2025

- Improved ED patient outcome prediction by 0.12 AUROC by tuning and deploying **LightGBM** models trained on historical patient data to a hospital dashboard, enabling better resource allocation for 100+ patients daily.
- Identified the optimal model architecture improving accuracy by 15% by benchmarking **PyTorch transformer** and **LLM-assisted LightGBM** models against baselines across millions of patient records.
- Improved ED physician confidence in model decisions by expanding a **React** dashboard and **AWS** serverless backend with SHAP-based explainability, making predictions transparent and actionable for physicians.

Rivian | *Software Engineer Intern* May 2024 – Dec 2024

- Built foundational tooling to validate Rivian's transition from **REST/GraphQL**-based communication to **Protocol Buffers**, maintaining 100% feature test coverage by developing Rivian's first **Python Protocol Buffer** pub/sub framework and NATS client tool to enable simulated cloud-based testing, reducing dependence on limited prototype hardware.
- Proposed and drove development of an **agentic** full-stack vehicle simulator built with **React** and **AWS** that let 100+ engineers and designers test mobile and cloud features, eliminating hours of authentication and CLI setup.
- Led mobile integration of the 2024 Halloween feature and multiple other features, shipping to 100,000+ customers across 7 major app releases by designing test strategy and driving cross-team alignment.

AWS Cloud Innovation Centre (UBC) | *Software Developer Co-op* Jan 2023 – Aug 2023

- Built a **Flutter/Dart** mobile app with a **TypeScript**-based **AWS** serverless backend to collect Parkinson's disease symptom data from thousands of patients annually with minimal discomfort, supporting clinical research teams.
- Improved balance assessment for rehabilitation patients using transformer-based ML on IMU sensor data, delivered via a **React** and **Flutter** UI with a **TypeScript** and **Python** AWS backend.
- Presented at the 2024 Interdisciplinary Conference in Rehabilitation Science BC (INCREASE BC).

PROJECTS

FinHog · *Agentic AI, TypeScript, Python, AWS, React* Sep 2025

- Won 1st place at HackMIT (1,500 participants) for Anthropic's Best Use of Claude, and 2nd place for Warp's (YC W23) Best Financial Visualization Agent.
- Built an autonomous **financial analytics agent** that analyzes financial data in real time, automatically generating and adapting visualizations to surface the most meaningful insights through various statistical analysis tools.

Vancouver Transit App · *Dart, Flutter, iOS, Android* Nov 2019 – Jan 2022

- Built a **Flutter**-based iOS and Android app providing real-time transit data via a custom GTFS **protobuf** decoder, reaching 1,000+ active users and 10,000+ installs across the Apple App Store and Google Play Store.

TECHNICAL SKILLS

Languages	C++, Java, Dart, TypeScript, Python, SQL, JavaScript, GraphQL, HTML/CSS
Frameworks & Libraries	Protocol Buffers, AWS CDK, Flutter, React, Node.js, Pandas, NumPy, JUnit
Tools & Platforms	AWS, PostgreSQL, MySQL, Docker, Git, CI/CD, NoSQL, Xcode, Android Studio, VS Code